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Skill 27.1 Exercise 1

Refer to the movies DataFrame below.

	movie	production_budget	domestic_gross	worldwide_gross	mpaa_rating	genre
0	Evan Almighty	175000000.0	100289690.0	174131329.0	PG	Comedy
1	Waterworld	175000000.0	88246220.0	264246220.0	PG-13	Action
2	King Arthur: Legend of the Sword	175000000.0	39175066.0	139950708.0	PG-13	Adventure
3	47 Ronin	175000000.0	38362475.0	151716815.0	PG-13	Action
4	Jurassic World: Fallen Kingdom	170000000.0	416769345.0	1304866322.0	PG-13	Action

For each variable in the *movies* DataFrame, indicate whether it is *quantitative* or *categorical*

How could you use the `describe()` method to display the summary statistics for all the columns in the *movies* DataFrame?

Skill 27.2 Exercise 1

Refer to the movies DataFrame below.

	movie	production_budget	domestic_gross	worldwide_gross	mpaa_rating	genre
0	Evan Almighty	175000000.0	100289690.0	174131329.0	PG	Comedy
1	Waterworld	175000000.0	88246220.0	264246220.0	PG-13	Action
2	King Arthur: Legend of the Sword	175000000.0	NaN	139950708.0	PG-13	Adventure
3	47 Ronin	NaN	38362475.0	151716815.0	PG-13	Action
4	Jurassic World: Fallen Kingdom	170000000.0	416769345.0	NaN	NaN	Action

Select the *production_budget* column and store it in a variable called *budget*.

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Write to code to answer the following,

- What is the data type of *production_budget*?
- How many missing values does it have?
- How many unique values does it have?
- What are the unique values?

Skill 27.3 Exercise 1

Refer to the movies DataFrame below which shows the first 5 rows of a larger dataset.

	movie	production_budget	domestic_gross	worldwide_gross	mpaa_rating	genre
0	Evan Almighty	175000000.0	100289690.0	174131329.0	PG	Comedy
1	Waterworld	175000000.0	88246220.0	264246220.0	PG-13	Action
2	King Arthur: Legend of the Sword	175000000.0	NaN	139950708.0	PG-13	Adventure
3	47 Ronin	NaN	38362475.0	151716815.0	PG-13	Action
4	Jurassic World: Fallen Kingdom	170000000.0	416769345.0	NaN	NaN	Action

Calculate the mean and median of *production_budget*.

The actual values of the mean and median are 33284743.23 and 20000000.0, respectively, what measure better represents a "typical" movie? Justify your answer.

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Determine the mode for the <i>mpaa_rating</i> . Why is the mode more appropriate than the mean or median here?

Skill 27.4 Exercise 1

Refer to the movies DataFrame below which shows the first 5 rows of a larger dataset.

	movie	production_budget	domestic_gross	worldwide_gross	mpaa_rating	genre
0	Evan Almighty	175000000.0	100289690.0	174131329.0	PG	Comedy
1	Waterworld	175000000.0	88246220.0	264246220.0	PG-13	Action
2	King Arthur: Legend of the Sword	175000000.0	NaN	139950708.0	PG-13	Adventure
3	47 Ronin	NaN	38362475.0	151716815.0	PG-13	Action
4	Jurassic World: Fallen Kingdom	170000000.0	416769345.0	NaN	NaN	Action

Calculate the min, max, and range of <i>production_budget</i> .

Calculate the IQR

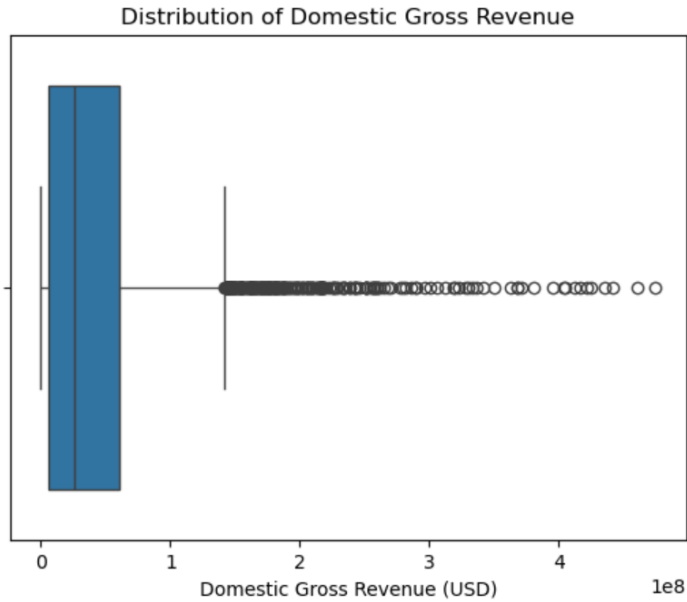
The actual values of the range and IQR are 74750000.0 and 36000000.0, respectively. How do extreme blockbuster movies affect each measure? Which statistic better describes the spread of <i>most</i> movies?

Calculate the standard deviation
The actual value of the standard deviation is 34892390.59. Based on this value and the previous values, is the data symmetric or skewed?

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Skill 27.5 Exercise 1

Below is a box blot of the movie data from the previous exercise.



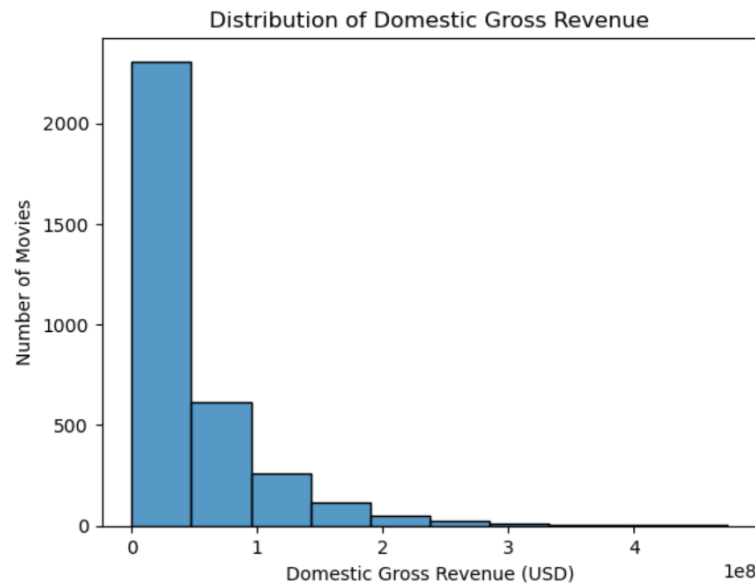
Label each of the following on the plot:

- Left whisker
- Right whisker
- Quantile 1, Quantile 2, and Quantile 3
- Outliers
- ICQ

Based on this boxplot, which measure of center and which measure of spread would best describe a typical movie? Explain why.

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Below is a histogram of the movie data from the previous exercise.



How would this histogram change if there were no extremely high-grossing movies?
Which measure of center and spread would become more appropriate?

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Skill 27.6 Exercise 1

Refer to the movies DataFrame below which shows the first 5 rows of a larger dataset.

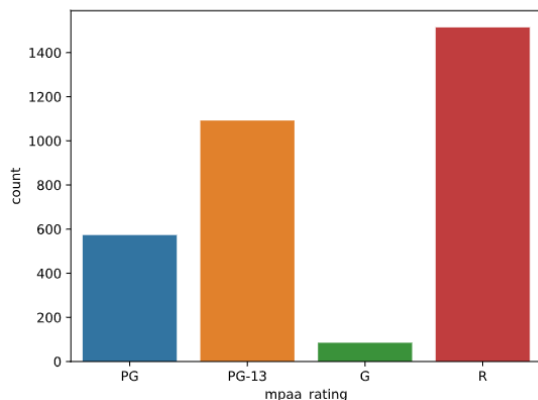
	movie	production_budget	domestic_gross	worldwide_gross	mpaa_rating	genre
0	Evan Almighty	175000000.0	100289690.0	174131329.0	PG	Comedy
1	Waterworld	175000000.0	88246220.0	264246220.0	PG-13	Action
2	King Arthur: Legend of the Sword	175000000.0	NaN	139950708.0	PG-13	Adventure
3	47 Ronin	NaN	38362475.0	151716815.0	PG-13	Action
4	Jurassic World: Fallen Kingdom	170000000.0	416769345.0	NaN	NaN	Action

Use the `value_counts()` method to count how many movies fall into each `mpaa_rating`. Describe the expected output.

Use the `normalize` attribute to convert the MPAA rating counts above to proportions.

Use `sns.countplot()` to create a bar chart of the counts.

Below is a bar chart which shows the counts for each movie in the `mpaa_rating` column.



What type of movie is most/least common?